

Project Design

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Project Design - Sistema de Gestión para Lavadero y Parqueadero

CRC Cards

Vehicle	
Responsibilities:	Collaborators:
Maintain vehicle data (license plate, type, current state).	WashingService
Register entry and exit for parking or washing services.	MonthlySubscription
Calculate total parking time for billing.	Agreement
Associate with monthly plans or corporate agreements.	Shift

Washer	
Responsibilities:	Collaborators:
Record washing services performed.	WashingService
Calculate daily bonuses based on completed washes.	Bonus
Manage vouchers and deductions linked to bonuses.	Voucher
	Shift

WashingService	
Responsibilities:	Collaborators:
Represent an individual car wash service.	Vehicle
Maintain service status (Pending, In Progress, Completed).	Washer
Log timestamps for each status change.	Shift
	Bonus



Calculate cost based on washing type.

Link to both the washer and the corresponding vehicle.

Shift	
Responsibilities:	Collaborators:
Manage vehicles, washes, and expenses for each shift.	Vehicle
Consolidate economic data (income, expenses, bonuses).	WashingService
Associate logs with the responsible operational admin.	Expense
Generate individual shift reports.	OperationalAdmin

OperationalAdmin	
Responsibilities:	Collaborators:
Register vehicle and wash entries.	Shift
Assign washers to services.	WashingService
Record and categorize shift expenses.	Washer
Generate performance reports per shift.	Expense

GlobalAdmin	
Responsibilities:	Collaborators:
Define base rates, bonus percentages, and business policies.	Rate
	Shift



Manage users (washers, operational admins).	FinancialReport
Configure free parking times per wash type.	Bonus
Analyze general metrics and consolidated reports.	
Delete or correct expense records.	

MonthlySubscription	
Responsibilities:	Collaborators:
Register and manage vehicles under a monthly plan.	Vehicle
Calculate expiration and alert the system.	GlobalAdmin
Apply discounts or adjustments based on business rules.	

Agreement	
Responsibilities:	Collaborators:
Link corporate clients with their vehicle fleets.	Vehicle
Define discounts by service or wash type.	GlobalAdmin
Generate consolidated monthly reports per company.	FinancialReport

Bonus	
Responsibilities:	Collaborators:
Calculate and accumulate daily/monthly washer bonuses.	Washer
Apply automatic deductions for vouchers.	Voucher



Associate bonuses with corresponding washer and period.	WashingService Shift
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Voucher	
Responsibilities:	Collaborators:
Register advances or loans made to washers. Calculate deduction amounts for each payment period. Maintain state (Active, Settled).	Washer Bonus

Expense	
Responsibilities:	Collaborators:
Register expenses by date and shift. Categorize expenses (supplies, maintenance, etc.). Associate expenses with either global or operational admins.	Shift OperationalAdmin GlobalAdmin

FinancialReport	
Responsibilities:	Collaborators:
Consolidate income, expense, and bonus data. Calculate business performance (Income – Expenses – Bonuses). Generate graphical and exportable reports.	Shift Bonus Expense GlobalAdmin

AuthService



Responsibilities:	Collaborators:
Authenticate users (global and operational admins). Handle password recovery via email. Manage roles and access permissions.	GlobalAdmin OperationalAdmin EmailService

WashingServiceHandler	
Responsibilities:	Collaborators:
Manage the full wash cycle (registration, assignment, status change). Calculate service times and costs. Record data for performance metrics.	WashingService Washer Vehicle Shift

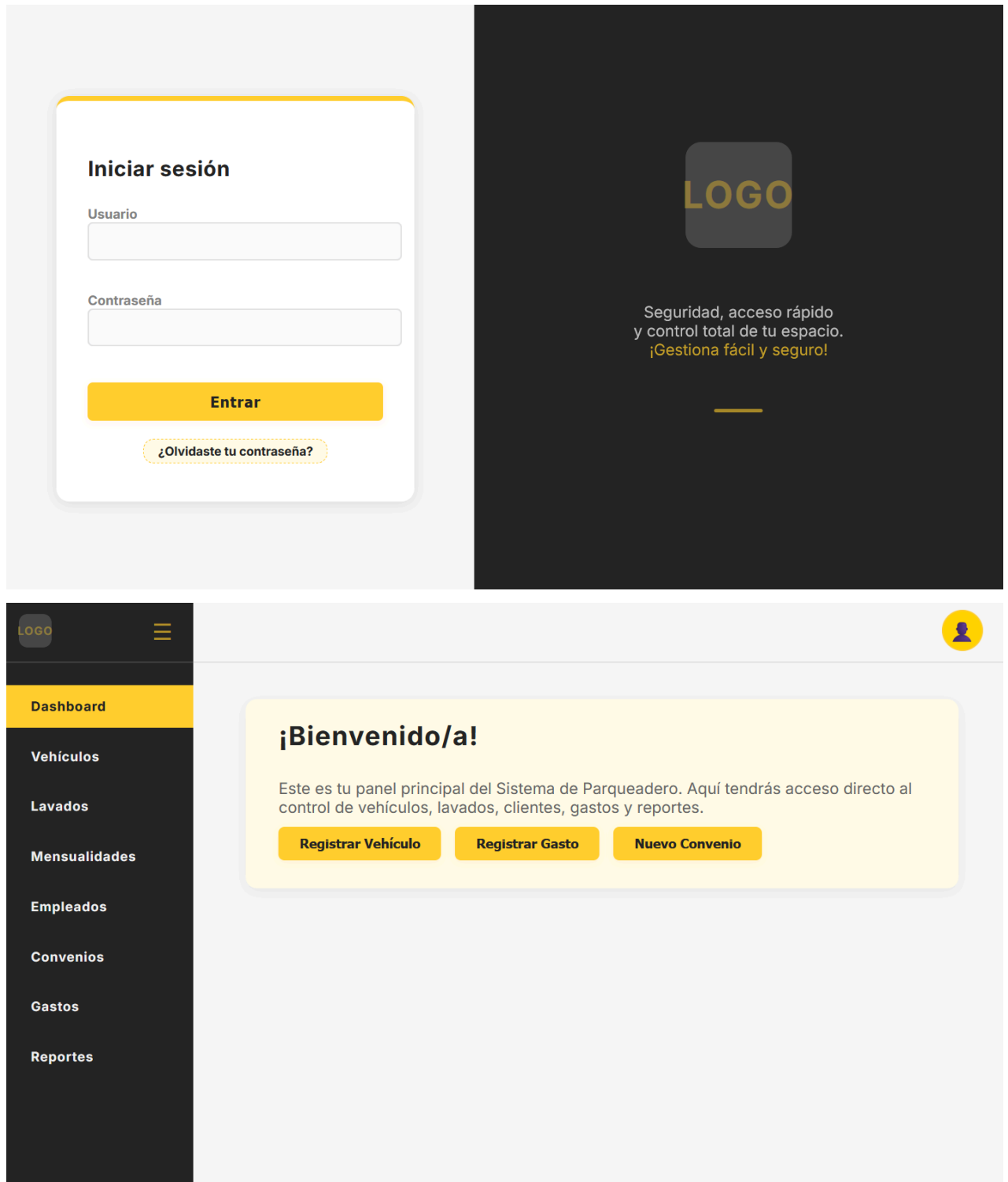
ParkingServiceHandler	
Responsibilities:	Collaborators:
Register vehicle entry and exit. Calculate parking fees based on time. Integrate free parking time logic with washing services.	Vehicle Shift OperationalAdmin

BonusService	
Responsibilities:	Collaborators:
Compute daily and monthly washer bonuses. Apply voucher deductions automatically. Generate individual performance reports.	Washer Bonus Voucher Shift



ReportService	
Responsibilities:	Collaborators:
Generate business metrics and graphical reports.	FinancialReport
Calculate overall performance and income vs. expense comparisons.	Expense
Consolidate data for dashboard visualization.	Bonus
	Shift

Mockups



LOGO

Dashboard

Vehículos

Lavados

Mensualidades

Empleados

Convenios

Gastos

Reportes

Registrar Vehículo

Buscar por placa... Buscar

Placa	Tipo	Cascos	Hora ingreso	Acciones
ABC124	Automóvil	-	09:23	Registrar salida Historial
JKL56A	Moto	2	10:01	Registrar salida Historial
LMN893	Camioneta	-	08:57	Registrar salida Historial

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Registrar Lavado

Buscar por placa... Buscar

Placa	Tipo	Lavador	Estado	Hora inicio	Acciones
JKL56A	Moto	Carlos P.	Pendiente	10:10	En proceso Terminar Detalle
LMN893	Camioneta	Lucía G.	En proceso	09:44	Terminar Detalle
XYZ235	Automóvil	--	Terminado	09:06	Detalle



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Reportes

Gestión de Mensualidades

Registrar Mensualidad

Buscar por placa o cliente...

Buscar

Placa	Cliente	Vigencia	Estado	Acciones
XYZ235	Juan Pérez	2025-12-05	Activa	Renovar Historial
JKL56A	Lucía Gómez	2025-11-13	Por vencer	Renovar Historial
LMN893	Empresa XYZ	2026-01-04	Activa	Renovar Historial

LOGO

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Reportes

Gestión de Convenios

Registrar Convenio

Buscar empresa o contacto...

Buscar

Empresa	Condiciones	Contacto	Acciones
EmpresaABC	Descuento 10%, flota 18 vehículos	Laura Rojas	Flotilla Editar Reportes
Transportes XYZ	Descuento 15%, crédito autorizado	Carlos Pardo	Flotilla Editar Reportes
SegurosSegura	Descuento 8%, flota 10 motos	Marta Niño	Flotilla Editar Reportes



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Reportes

Gestión de Empleados

Registrar Empleado

Buscar por nombre o cargo...

Buscar

Nombre	Cargo	Bono %	Servicios	Estado	Acciones
Juan Pérez	Lavador	10%	32	Activo	Editar Detalle Eliminar
Lucía Gómez	Lavador	15%	28	Activo	Editar Detalle Eliminar
Rubén Ramirez	Administrador	0%	-	Inactivo	Editar Eliminar
Mario Cuellar	Lavador	10%	14	Bloqueado	Editar Eliminar

LOGO

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Reportes

Reportes y Exportación

Exportar CSV

Fecha inicio

dd / mm / aaaa

Fin

dd / mm / aaaa

Tipo

Todos

Filtrar

Ingresos\$240.000

Gastos\$120.000

Salidas70

Fecha	Tipo	Descripción	Monto	Usuario	Acciones
2025-11-11	Ingreso	Ingreso de automóvil	\$6.000	Juan Pérez	Detalle
2025-11-11	Gasto	Compra de jabón	\$18.000	Lucía Gómez	Detalle
2025-11-11	Lavado	Lavado Premium	\$10.000	Carlos Pardo	Detalle

Registrar Vehículo

Placa

Tipo

Cascos (sólo moto)

Registrar

Registrar Empleado

Nombre

Cargo

Bono (%)

Estado

Registrar

Registrar Lavado

Placa vehículo

Tipo de lavado

Lavador (opcional)

Registrar

Registrar Gasto

Tipo

Descripción

Monto

Usuario

Registrar

Architecture Diagram

carwash_architecture_c4.svg

Class Diagram

umldiagram.svg

Relational Database Model:

RelationalDatabaseModel.svg

Business Model Processes

WashingServiceProcess.svg

